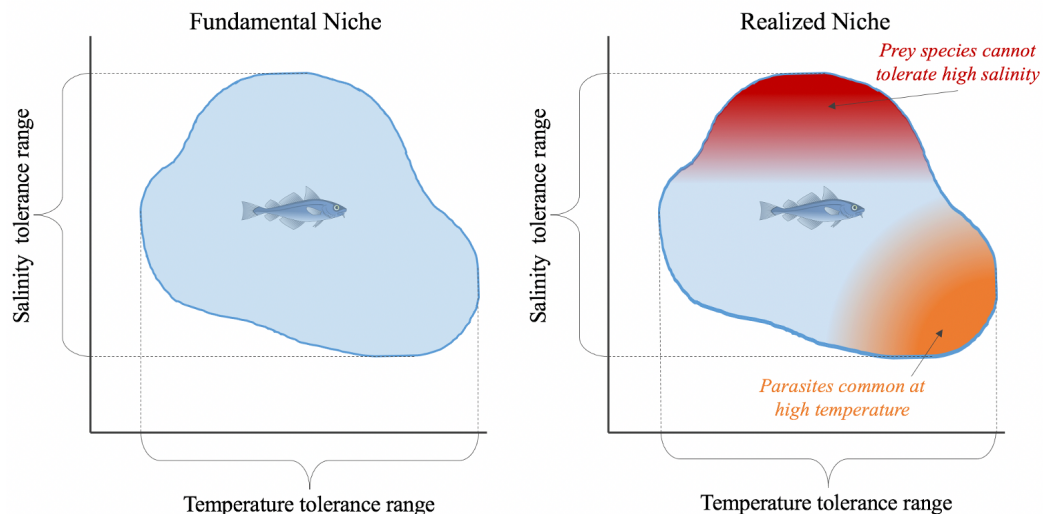
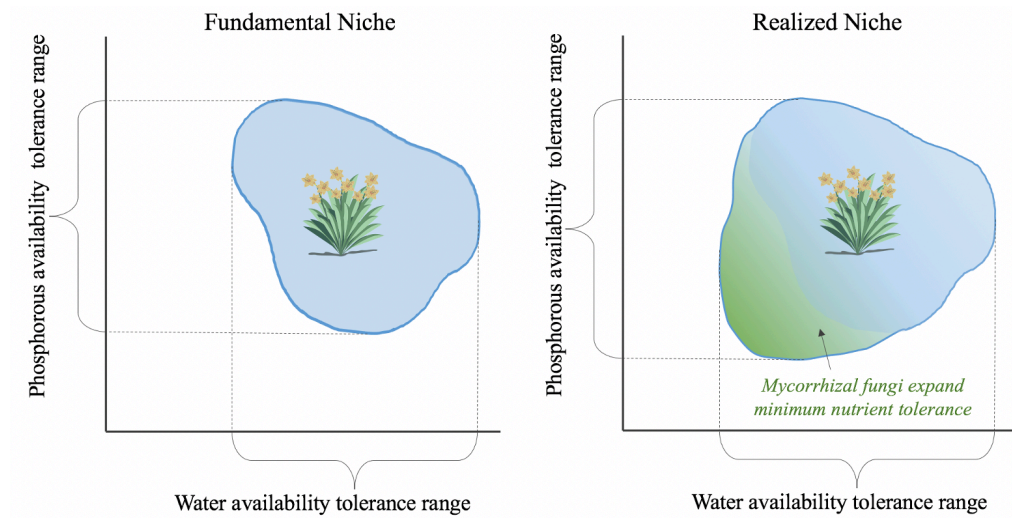


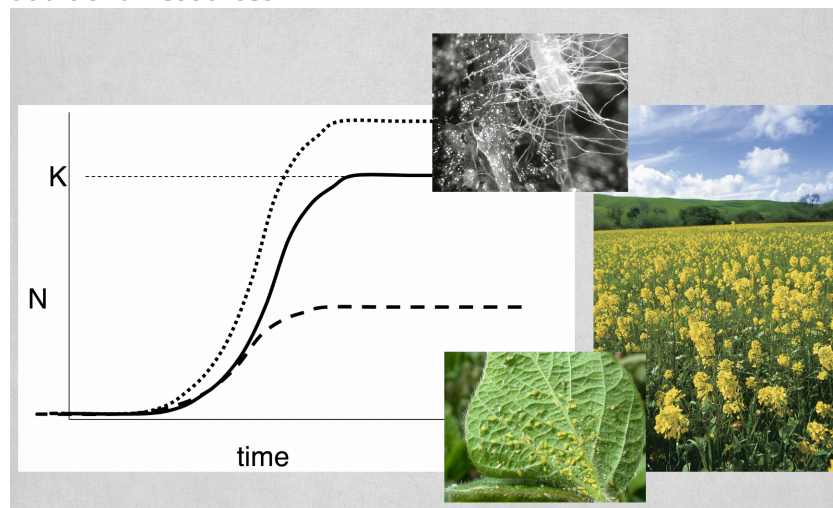
Species Interactions Study Guide KEY

1. Blank1: realized; blank2: biotic
2. A, B, and C would be considered symbiotic
3. 1: mutualism; 2: commensalism; 3: exploitation; 4: amensalism; 5: competition
4.
 - a. d_p
 - b. p
 - c. r
 - d. c
 - e. this term is not represented in the Lotka-Volterra equations
5.
 - a. Competitive exclusion
 - b. Resource partitioning
 - c. Competitive exclusion
6. A, B, C, and D
7. A
8. B
9. Blank1: competition; blank2: minus; blank3: minus
10. Fundamental niche: **abiotic** conditions in which a species can survive and reproduce (determined by physiological tolerance)
Realized niche: the **biotic** interactions that determine where a species can survive and reproduce (determined by species interactions)
Carrying capacity: the maximum number of organisms of a given species that a habitat can support (determined by resource availability)
 - a. Examples from lecture:





- b. Examples from lecture: parasites or pathogens (such as aphids in the photo) might reduce carrying capacity if individuals must be sufficiently spread out in the environment to not infect one another; mutualistic relationships (such as mycorrhizae in the photo) may increase the carrying capacity through increased efficiency of resources use, or access to additional resources.



11. A **keystone species** is one whose impact on the ecosystem (especially in terms of species richness or evenness) is high despite that species being relatively rare in the ecosystem. The example we discussed in class was seastars, which promote species richness in tidepools by consuming a variety of species, which then keeps each of the prey populations low enough that none outcompete the others. A **foundation species** is one whose presence (especially in terms of their physical bodies) makes up the habitat for the other species. A classic example is coral reefs, in which the coral itself is the structure of the habitat, and forms the physical spaces that other species inhabit (caves, crevices between corals, etc). Foundation species must be common in the ecosystem, and so a foundation species can't really be a keystone species (because keystone species are by definition rare). An **ecosystem engineer** is one who physically alters or modifies the habitat in significant ways. A classic example is beavers, which alter the flow of rivers with

their dams.

12. Answers:

- A. True
- B. True
- C. True
- D. True
- E. False

13. The competitive exclusion principle states that two species that require the same limiting nutrient cannot coexist; one species will be better able to access or use that resource and will eventually drive the other to extinction in the shared habitat. Organisms can avoid this outcome by limiting competition between themselves and other species through 1) resource partitioning: in which the limiting resources is partitioned and one species accesses only one component of the resources and the other access a different component. Classic examples include physically separating in the habitat (lizard and bird examples in lecture) or using different 'categories' within the resource (the bat example of types and size of insects). Or through 2) character displacement, which is a form of resource partitioning that is coupled with a morphological adaptation for accessing a component of the resource. Classic examples include alterations of beak shape and size to adapt to specific food sources in finches and honeycreepers

14.

- a. In both species, the population size is driven primarily by the size of the other species. When lynx are numerous, hare population size will decline which will then drive reductions in lynx populations. When lynx are not numerous, hare population size will increase, which will then drive increases in the lynx population.
- b. There is a bit of a lag between the change in one population and the resulting change in the other. The easiest way to think about why this lag occurs is that when prey populations are high, it takes some time for the predators to consume those additional prey and convert them into new offspring. This time delay produces the lag in the max and min population sizes.

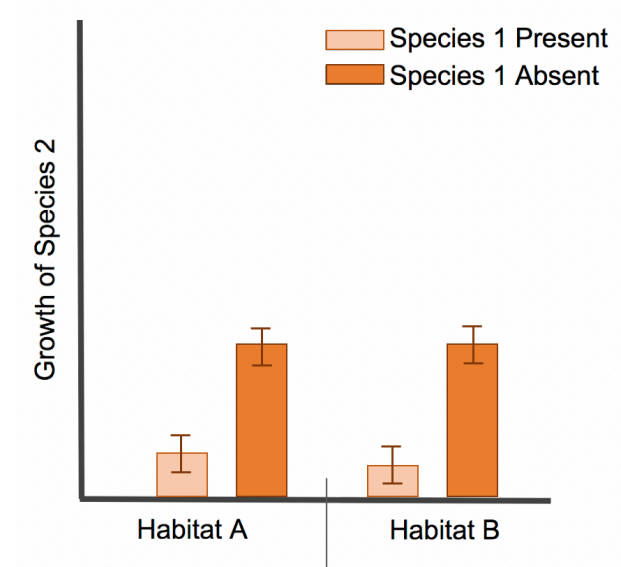
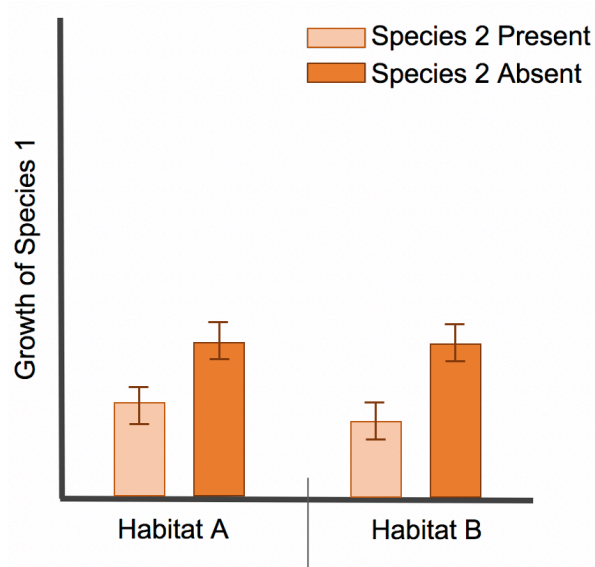
15.

- a. The equation for prey includes the variable P (pop size of predators) and the equation for predators includes the variable V (pop size of victims/prey).
- b. cpV is equivalent to r for the predator population. The victim population size, predation rate, and conversion efficiency all contribute to how quickly the predator population can increase.
- c. Lotka and Volterra used the exponential equation instead of the logistic equation because they believed that the primary driving factor of population sizes in predators and prey was the other's population size and that neither would ever reach the carrying capacity of the ecosystem, making K irrelevant in these equations.

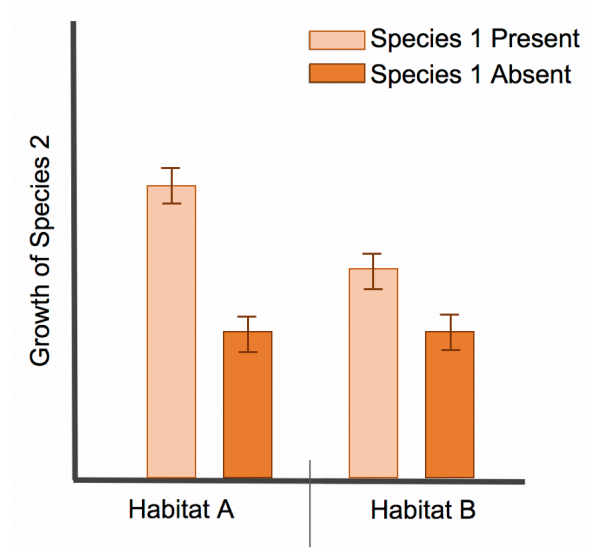
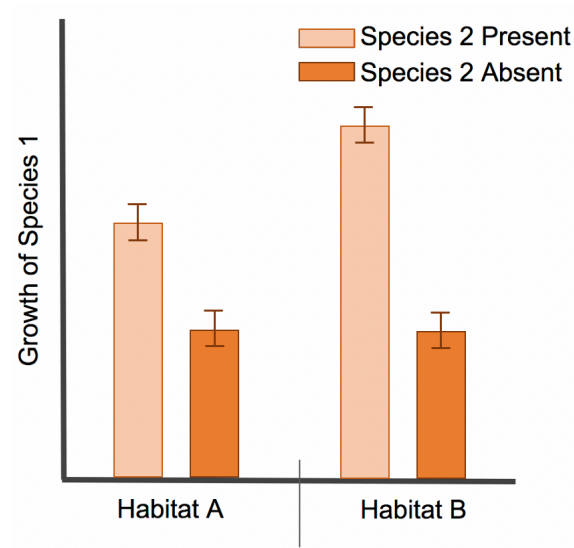
16. As we saw in the orange mite experiment, if a predator is too efficient then it drives its prey extinct and then also dies out itself. The same is true for a pathogen – if it kills its host before the host can transmit it to another individual, then the pathogen dies out too. Consequently, there is

some natural selection for reduced virulence or efficiency, because if you are too efficient, you use up your resources (hosts) too quickly and then also die.

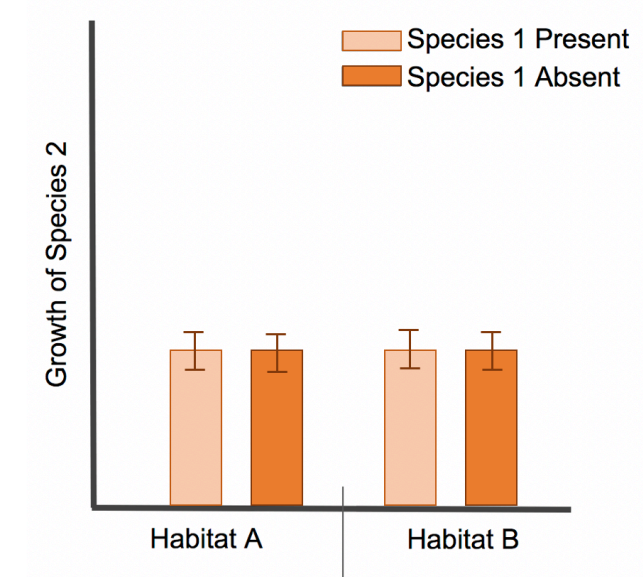
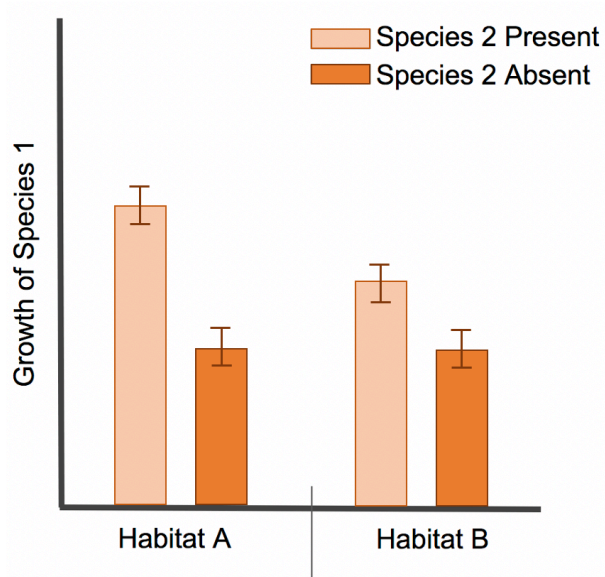
17. Infection by pathogens is density dependent, meaning that individuals are more likely to contract the disease if they are coming into frequent contact with other individuals who may be infected. Herd immunity operates by using vaccination to reduce the number of individuals who could be infected, which effectively reduces the density of potentially infected individuals in the population to a low enough level that susceptible individuals are unlikely to come into contact with them.
18. Measles is one of the most contagious diseases found in the United States (having the highest reported R_0 value) and rubella is nearly as high. This means that these illnesses require a very high percentage of vaccinated individuals (over 90% for the high end of R_0 estimates for measles) in order to protect unvaccinated individuals from illnesses. While this is not shown in the graph (and therefore would not be expected as part of your answer on an exam), you may also know that severe cases of measles and rubella (including mortality) are much higher than that of the flu. Consequently, a high R_0 value, and a higher potential for severe impacts of contracting the illnesses have made measles and rubella vaccines a requirement for public school children, while the flu vaccine is optional.
19. Examples of each type are below – note that in most cases, there are multiple ways to draw the graphs that correctly illustrate each relationship
Competition: both species are negatively impacted by the presence of the other species in both habitats (though the magnitude of the impact may not be identical on both species or in both habitats)



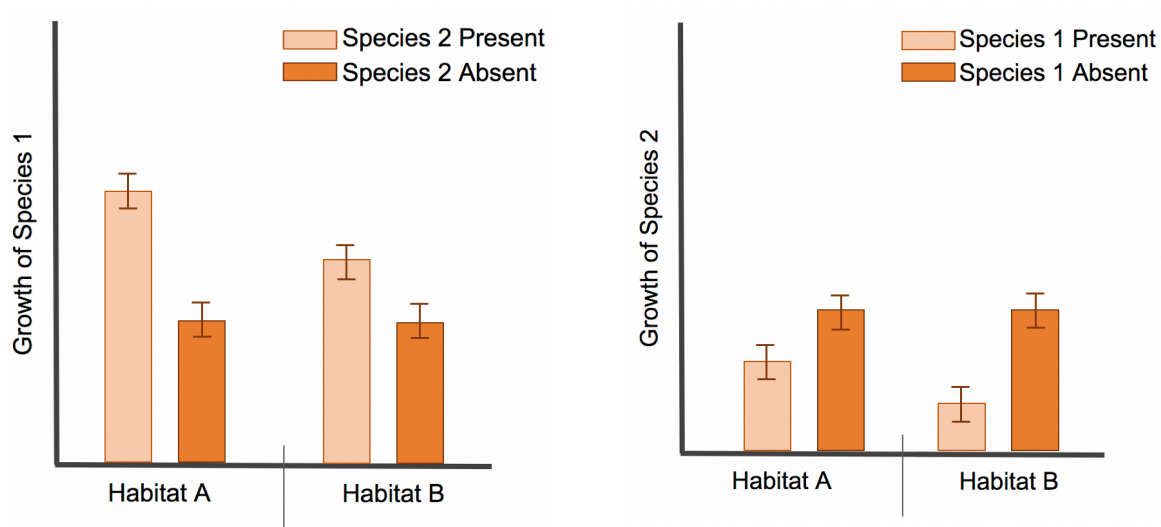
Mutualism: both species are positively impacted by the presence of the other species in both habitats (though the magnitude of the impact may differ between species or habitats)



Commensalism/Facilitation: one species is positively impacted by the presence of the other species in both habitats (though the magnitude of impact may differ between habitats). The other species is not significantly impacted.



Parasitism/Predation/Herbivory: one species is positively impacted by the presence of the other species in both habitats. The other species is negatively impacted by the presence of the other species. The magnitude of impacts may differ across habitats. In this example, Species 1 is the predator/herbivore/parasite and Species 2 is the prey/plant/host



20.

- Character displacement occurs when two species that have similar resource usages begin to compete with one another for that resource. Natural selection can separate their resource usage by benefitting phenotypes at the extreme of each species, which serves to reinforce the portions of resource usage that do not overlap (Laci showed graphs in lecture that illustrate how this occurs). Character displacement is similar to resource partitioning except that character displacement involves a shift in a morphological trait that relates to the resource usage (for example, beak shape). While this process actually occurs within the competing populations through time, we often are not tracking the process as it occurs, so how we generally find evidence that character displacement has occurred is by comparing trait usage in two species when they occur allopatrically (not in contact with each other) and when they occur sympatrically (in contact with one another). If the morphological trait and its resource usage overlap a lot in allopatry, but have diverged and do not overlap at all or very much in sympatry, then we have evidence of character displacement (Laci showed example graphs of this in class).
- Trait matching occurs when two species evolve in relation to each other traits in a way that deepens or tightens a mutualistic interaction. For example, flower traits (color, scent, shape) may evolve in order to more strongly attract a particular pollinator species at the same time that the pollinator's traits are evolving to more efficiently access that flower species' nectar (such as the moth/orchid example Laci discussed in class). The evidence for this type of interaction often relates to the specificity of the interaction (in the moth/orchid case, specificity would mean the moth primarily or only pollinates this flower and the orchid is primarily or only pollinated by the moth and not by other species) and changes in phenotypes through time that tailor the morphology, behavior, etc of each species to the morphology, behavior, etc of the other. A key point here is that both species are benefitted by the interaction and selection works to tailor the interaction more positively or more effectively.
- An evolutionary 'arms race' occurs when predator and prey, or herbivore and plant, or parasite and host traits evolve antagonistically, meaning that prey/plant/host traits are

selected to protect that individual from the predator/herbivore/parasite while the traits of the predator/herbivore/parasite are selected to make that individual more effective at consuming the prey/plant/host. Often the evidence we see of this kind of pattern is very strong defenses in the prey/plant/host (such as high degrees of toxicity or defensive chemical compounds, etc) and high degrees of resistance or tolerance to the defenses in the predator/herbivore/parasite. These defenses and resistances were selected for through time: when the two species came into contact, prey/plants/hosts with some degree of defense were benefitted and selected for predators/herbivores/parasites with some degree of tolerance to that defense which selected for prey/plants/hosts with some degree of defense, etc etc